

REVIEW OF PREVIOUS LECTURE

Estimator $\hat{\theta}_n$:

$$Bias(\hat{\theta}_n) = E(\hat{\theta}_n - \theta) = E(\hat{\theta}_n) - \theta$$

$$Var(\hat{\theta}_n) = E[(\hat{\theta}_n - E(\hat{\theta}_n))^2]$$

Quality of $\hat{\theta}_n$:

$$\text{mean squared error (mse)} = E[(\hat{\theta}_n - \theta)^2]$$

$$mse = bias^2(\hat{\theta}_n) + Var(\hat{\theta}_n)$$

$X_1, \dots, X_n$ iid $\Rightarrow$ sample mean is an unbiased estimator of μ .

UNBIASED ESTIMATION OF VARIANCE

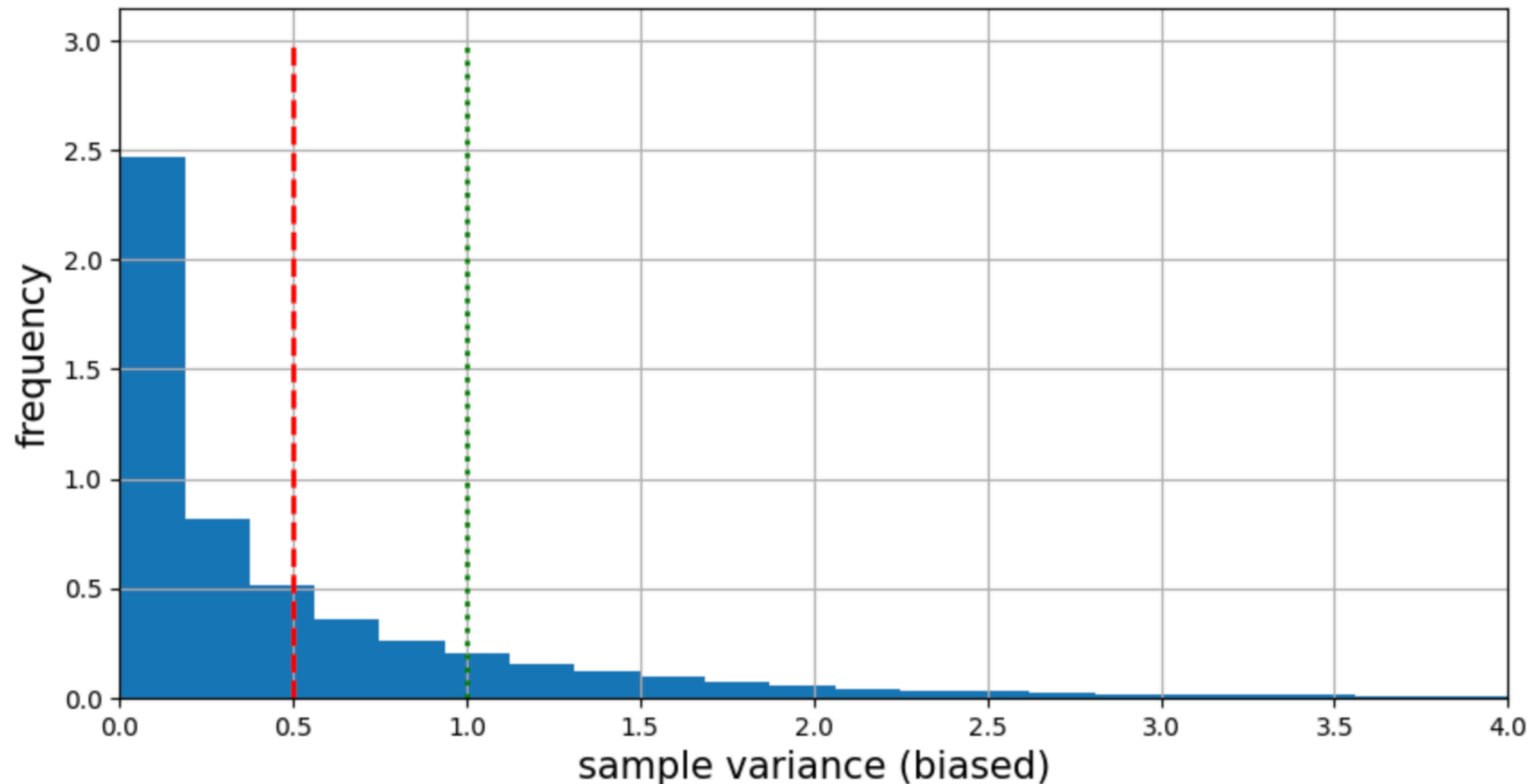
Reminder: $Var(X) = \sigma^2 = E(X - \mu_x)^2$

Estimate σ^2 :

1st try, plug-in estimate: $\hat{\sigma}_0^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$

UNBIASED ESTIMATION OF VARIANCE

```
n=2
s = 100000
X = np.random.normal(0,1,[n,s])
# ddof is 0(1) for dividing by n (n-1)
svar_b = np.var(X,axis=0,ddof=0)
mean_svar_b = np.mean(svar_b)
```



$$\widehat{\sigma_0}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

UNBIASED ESTIMATION OF VARIANCE

$$\widehat{\sigma}_0^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \text{ is biased.}$$

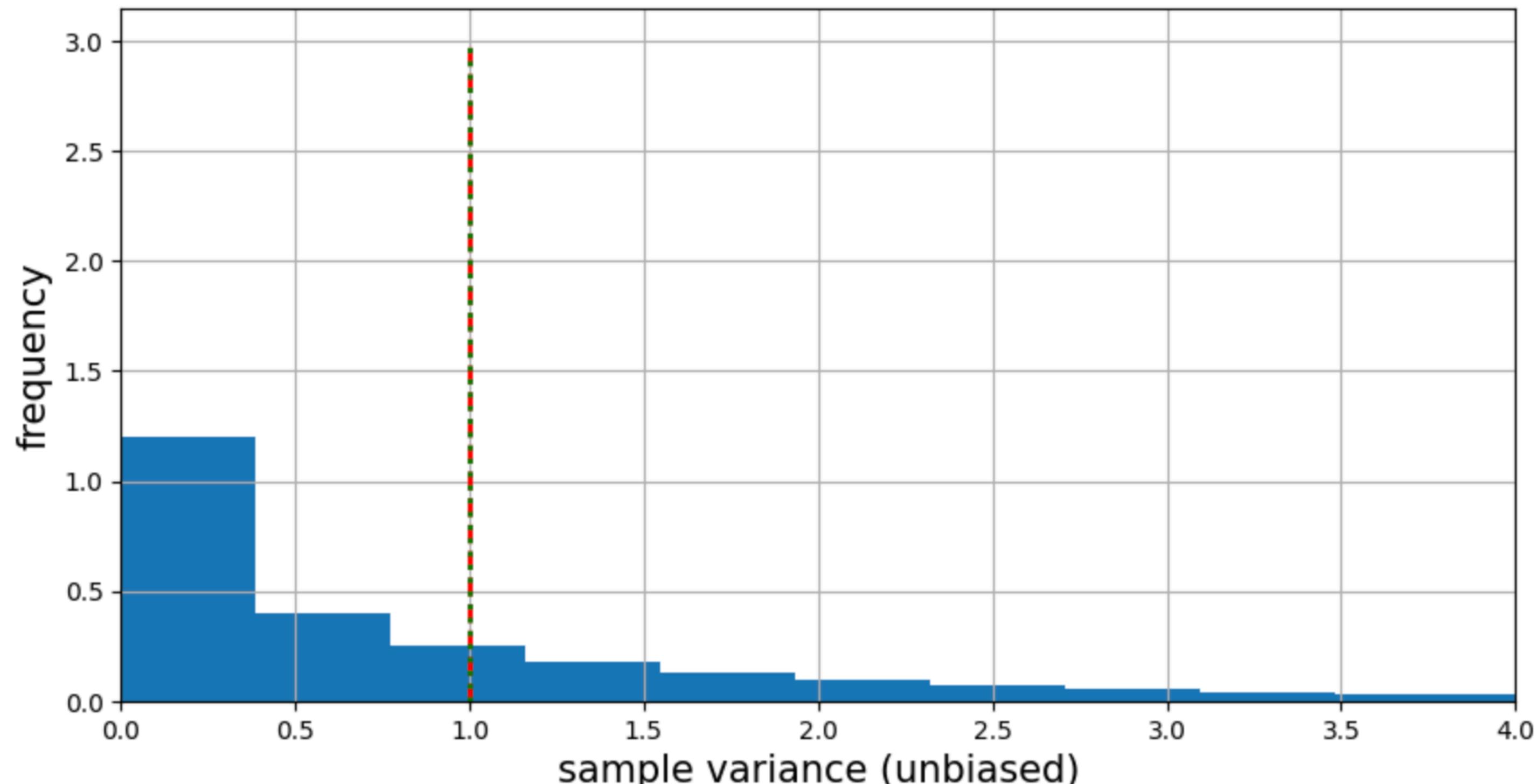
Theorem: $E(\widehat{\sigma}_0^2) = \frac{n-1}{n} \cdot \sigma^2$ (Check textbook for details)

Consider $\widehat{\sigma}_1^2 = \frac{n}{n-1} \cdot \widehat{\sigma}_0^2$:

$$\widehat{\sigma}_1^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \text{ is unbiased.}$$

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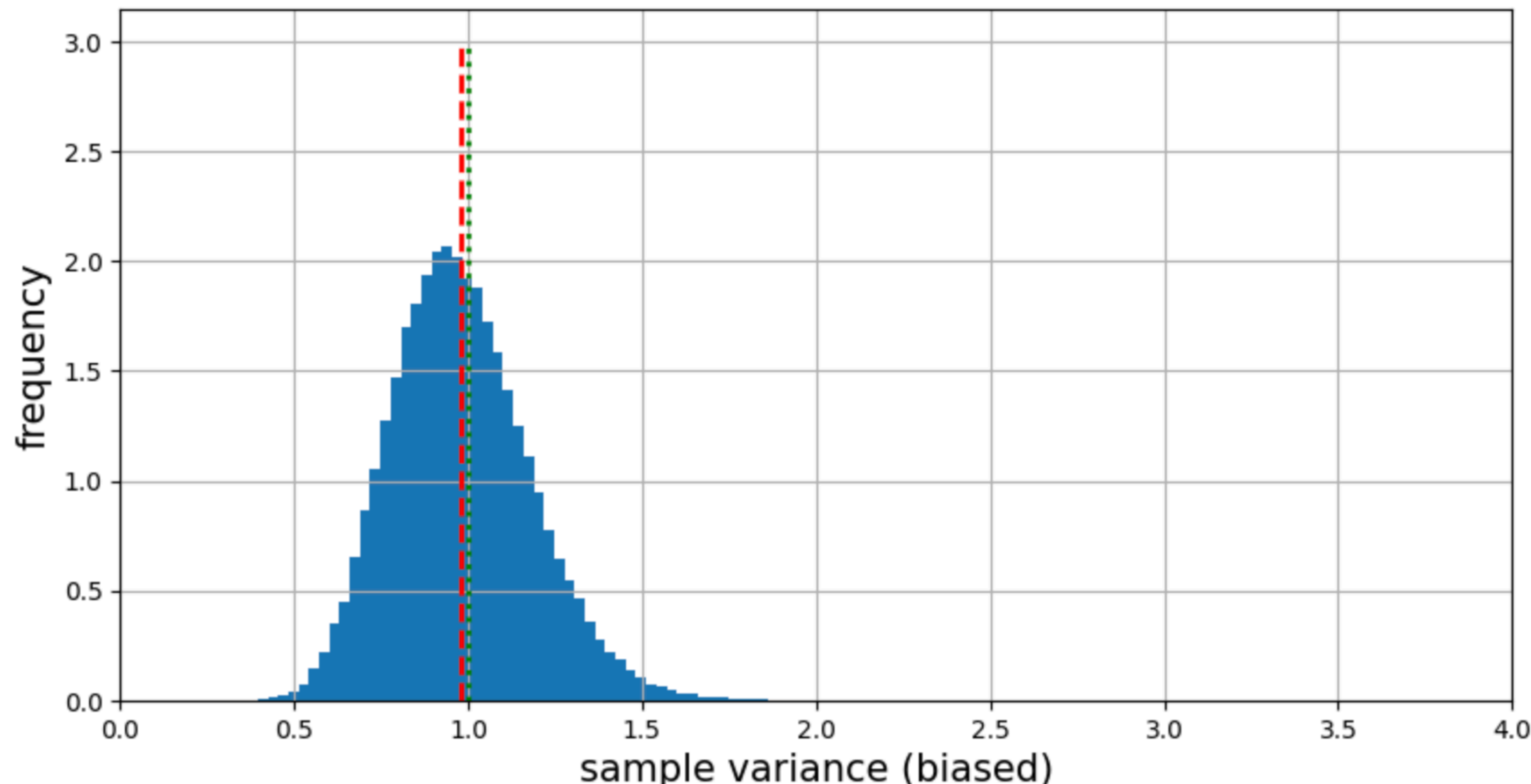
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$$\hat{\sigma}_1^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

UNBIASED ESTIMATION OF VARIANCE

```
n=50
s = 100000
X = np.random.normal(0,1,[n,s])
# ddof is 0(1) for dividing by n (n-1)
svar_b = np.var(X,axis=0,ddof=0)
mean_svar_b = np.mean(svar_b)
```



$$\widehat{\sigma}_0^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

For large n :

$$\widehat{\sigma}_0^2 \approx \widehat{\sigma}_1^2$$

MAXIMUM LIKELIHOOD ESTIMATORS

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$$0.75^3 0.25^1$$

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$\Rightarrow$ Our best estimate would be $\hat{p} = 0.75$.

MAXIMUM LIKELIHOOD ESTIMATORS

Likelihood function: Joint density of the data as a function of θ .

More specifically, let $X_1, \dots, X_n$ iid with pdf $f(x; \theta)$:

$$L(\theta) = \prod_{i=1}^n f(X_i; \theta)$$

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Maximizing log-likelihood is easier; summation rather than product

Simpler Goal: Find parameter θ that maximizes the log-likelihood.

$$\text{Solve for } \theta \text{ in } \frac{d \log L(\theta)}{d\theta} = 0$$

MAXIMUM LIKELIHOOD ESTIMATORS

Example: $X_1, \dots, X_n \sim \text{Bernoulli}(\theta)$. How to estimate unknown θ ?

$$L(\theta) = \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} \quad \text{e.g. with } 1, 0, 1, 1 \text{ the probability } \theta \cdot (1 - \theta) \cdot \theta \cdot \theta$$

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$$= \log \theta \sum_{i=1}^n x_i + \log (1 - \theta) (n - \sum_{i=1}^n x_i) \quad \text{Derivative wr.t. } \theta \text{ equal to 0 gives}$$
$$\theta = \bar{X}_n$$

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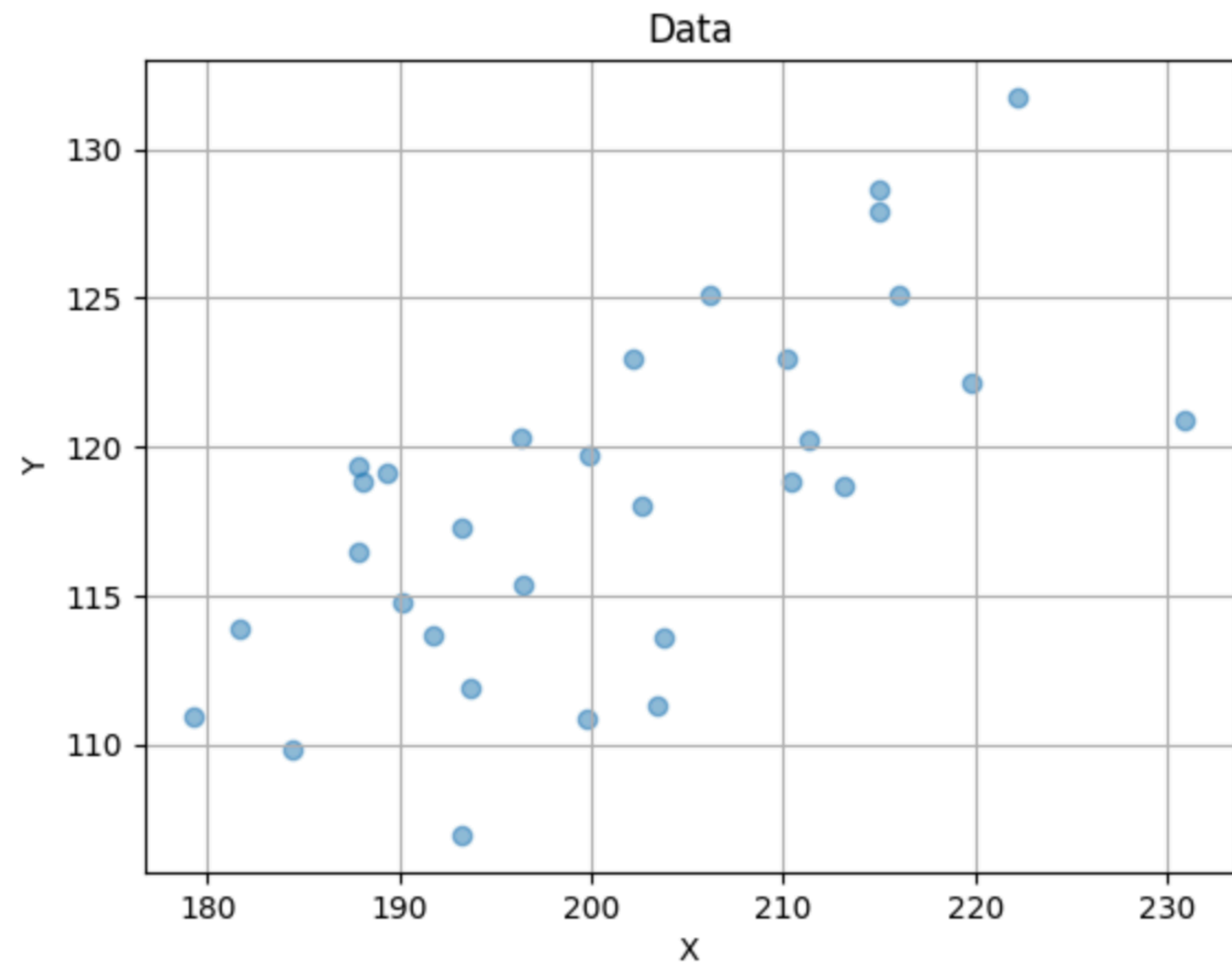
Same estimate as the plug-in estimator: sample mean.

Bias is 0.

Derivative wr.t. θ equal to 0 gives

$$\theta = \bar{X}_n$$

SIMULATION APPROXIMATION OF BOOTSTRAP ESTIMATES



PearsonRResult(statistic=0.6649024611769977, pvalue=4.498954341386631e-05)

X, Y have bivariate joint distribution.

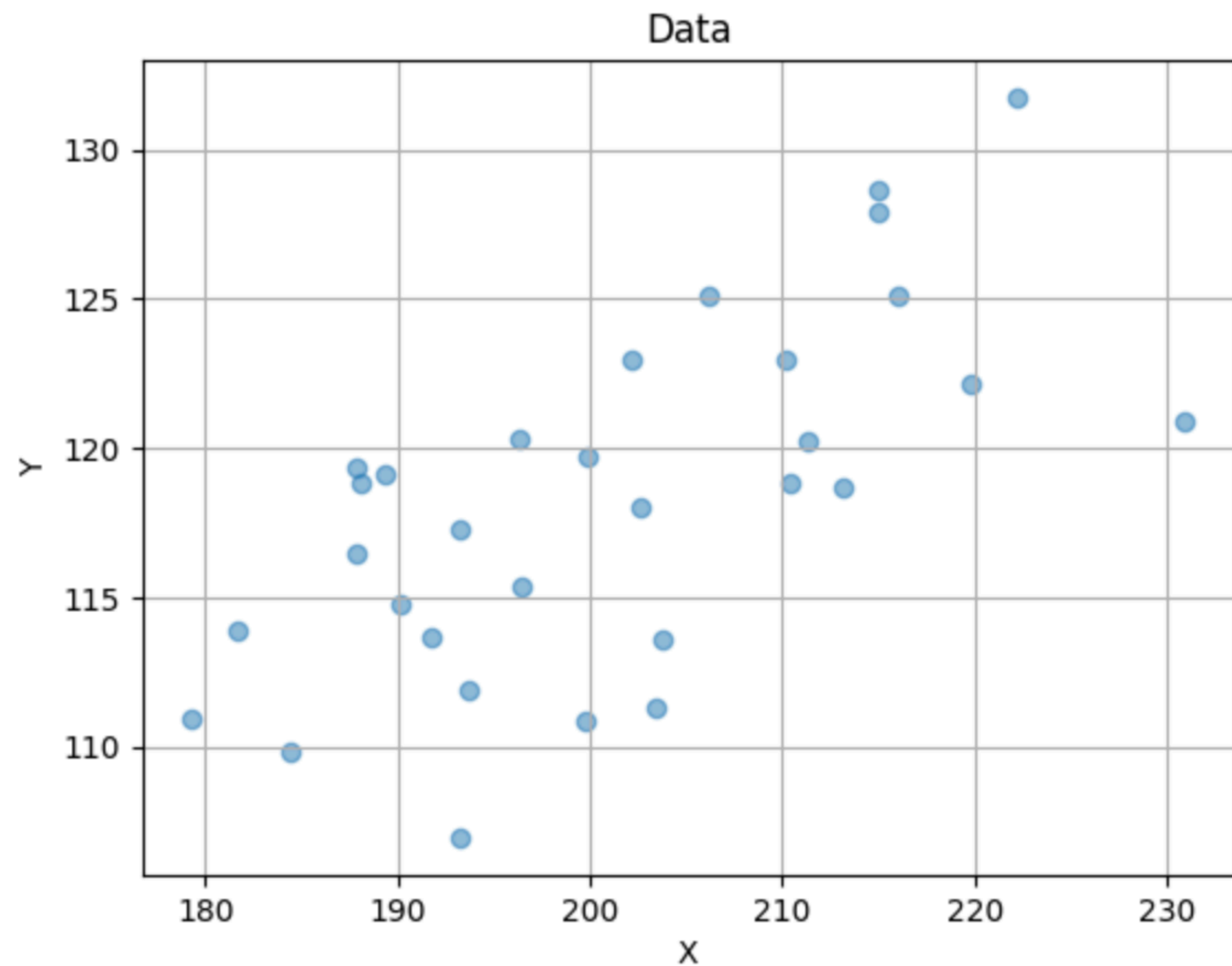
We observe a random sample (Data)

Sample Correlation:

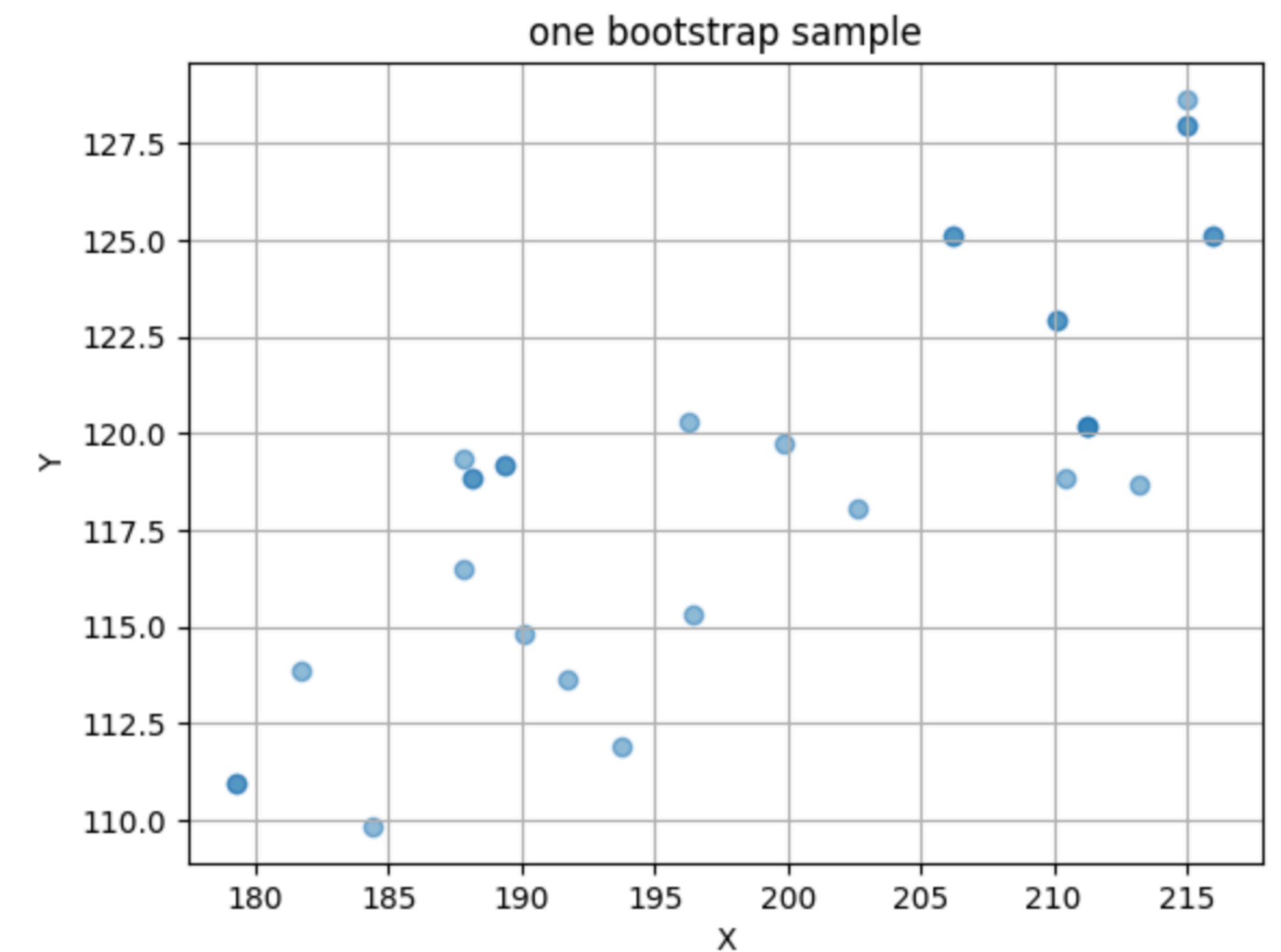
$$R = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}$$

Say we're interested in bias of R in estimating ρ .

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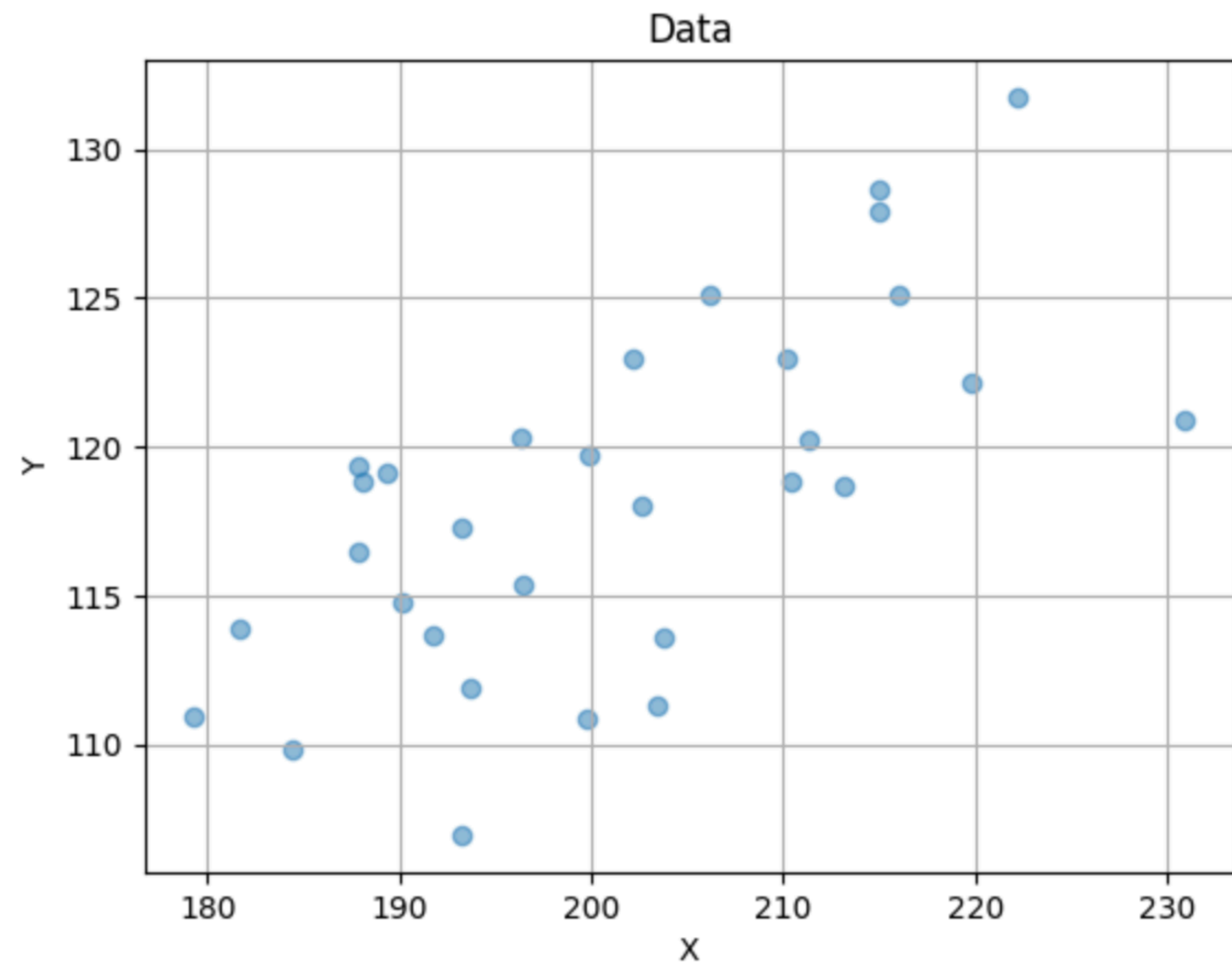
PearsonRResult(statistic=0.8015767674698848, pvalue=5.992620251964453e-08)

A bootstrap sample

Sample data with replacement

```
n_r = data.shape[0]  
rand_i = np.random.choice(n_r, size=n_r, replace=True)  
new_b = data[rand_i]
```

SIMULATION APPROXIMATION OF BOOTSTRAP ESTIMATES



PearsonRResult(statistic=0.6649024611769977, pvalue=4.498954341386631e-05)

```
n_samples=1000
n_r = data.shape[0]
diff_sum = 0
for _ in range(n_samples):
    rand_i = np.random.choice(n_r, size=n_r, replace=True)
    new_b = data[rand_i]
    corr = scipy.stats.pearsonr(new_b[:, 0], new_b[:, 1])
    diff_sum += corr[0]-corr_data
diff_sum/n_samples
```

0.0038447409039712647

R^i : Correlation of i^{th} bootstrap sample

Bias of sample correlation: Avg of $R^i - R$ to estimate mean of $R - \rho$

CONFIDENCE INTERVALS

Can apply to any parameter but most common is mean μ .

Have seen: $\bar{X}_n$ is a sensible point estimator for mean μ .

Precise estimate, like $\bar{X}_n = 4$

but how much confidence do we have that $\mu = 4$?

Confidence interval:

Make statements like

with 95 % confidence $\mu \in (\bar{X}_n - m, \bar{X}_n + m)$

CONFIDENCE INTERVALS

For μ with known σ :

Reminder 1:
$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \sim N(0,1)$$

Reminder 2:

$$p = P(-k \leq Z \leq k) \Rightarrow k = \Phi^{-1}\left(\frac{1+p}{2}\right)$$

CONFIDENCE INTERVALS

Given p for desired confidence find interval for μ using $\bar{X}_n$.

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Step 1: Find $k = \Phi^{-1}\left(\frac{1+p}{2}\right)$

Step 2: Find interval $\left(\bar{X}_n - \frac{k\sigma}{\sqrt{n}}, \bar{X}_n + \frac{k\sigma}{\sqrt{n}}\right)$



Margin of error